

Performance Testing

Date	4 July 2026
Team ID	
Project Name	Personalized Networking Assistant
Maximum Marks	5 Marks

Step 1: Testing Overview

Field	Details
Testing Tool Used	Manual testing via FastAPI Swagger UI and the Streamlit UI (no automated load-testing tool used — see note above)
Type of Testing	Manual response-time observation (informal)
Target Module / API	/analyze-event, /fact-check, /generate-conversation
Test Environment	Local (localhost, single machine, CPU-only inference)
Test Date	03 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1.	/analyze-event with a ~150-word event description	1	2–3	Themes returned within a few seconds
2.	/fact-check with a single topic query	1	1-2	Wikipedia summary returned, or graceful fallback if not found
3.	/generate-conversation full pipeline (theme extraction + fact retrieval + grounded generation)	1	10-20	Themes, facts, and 3 starters returned without error
4.	/generate-conversation with an empty/invalid interests field	1	Immediate	Graceful validation error, no crash

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
1.	Response Time (Avg)	Less than 2 seconds	~10–15 seconds for /generate-conversation	Fail (against generic target)	Target is a typical web-API default; not realistic for local CPU inference of two transformer models. See Observations below
2.	Response Time (Max)	Less than 5 seconds	~20 seconds observed	Fail (against generic target)	Same reasoning as above
3.	Throughput (Req/sec)	Not set	Not measured	N/A	Single-user manual testing only; no concurrent-load tool used, per scope
4.	Error Rate	Less than 1%	0% across manual test runs	Pass	No crashes or unhandled exceptions observed during testing
5.	CPU Utilization	Less than 80%	Not measured	N/A	No profiling tool used — out of scope for this build
6.	Memory Utilization	Less than 80%	Not measured	N/A	Same as above

Step 4: Observations & Analysis

Key Findings: The pipeline works correctly and reliably every time it was tested, but response time for /generate-conversation (10–20s) is well above conventional web-API targets. This is expected and inherent to running two transformer models (DistilBERT and GPT-2) sequentially on CPU, plus a live external call to Wikipedia — not a defect.

Bottlenecks Identified:

1. Theme extraction, fact retrieval, and generation run sequentially rather than in parallel.
2. GPT-2 generation (num_return_sequences=3) is the single largest time cost.
3. No caching. Identical Wikipedia queries are refetched on every request rather than cached.

Optimization Steps Taken: None implemented in this build, given the single-user local-demo scope, where response time was not a graded priority relative to correctness.

Step 5: Screenshots/Evidence

Personalized Networking Assistant 0.4.0 OAS 3.1

/openapi.json

default

POST

/analyze-event Analyze Event

POST

/fact-check Fact Check

POST

/generate-conversation Generate Conversation

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value

Schema

```
{
  "description": "A tech meetup exploring how artificial intelligence is being applied to renewable energy grids and smart city infrastructure.",
  "interests": ["machine learning", "climate tech", "urban planning"],
  "bio": "First-year Computer Science student interested in machine learning, with a growing interest in how AI can support sustainability efforts."
}
```

Execute

LOADING

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/generate-conversation' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "description": "A tech meetup exploring how artificial intelligence is being applied to renewable energy grids and smart city infrastructure.",
    "interests": ["machine learning", "climate tech", "urban planning"],
    "bio": "First-year Computer Science student interested in machine learning, with a growing interest in how AI can support sustainability efforts."
  }'
```

Request URL

http://127.0.0.1:8000/generate-conversation

Server response

Code

Details

200

Response body

```
{
  "topics": [
    "artificial intelligence",
    "technology",
    "entrepreneurship"
  ],
  "facts": [
    "Artificial intelligence (AI) is the capability of computational systems to perform tasks typically associated with human intelligence, such as learning, reasoning, problem-solving, percepti",
    "on, and decision-making. It is a field of research in engineering, mathematics and computer science that develops and studies methods and software that enable machines to perceive their environm",
    "ent and use learning and intelligence to take actions that maximize their chances of achieving defined goals.",
    "Technology is the application of conceptual knowledge to achieve practical goals, especially in a reproducible way. The word technology can also mean the products resulting from such effort",
    "s, including both tangible tools such as utensils or machines, and intangible ones such as software. Technology plays a critical role in science, engineering, and everyday life.",
    "Entrepreneurship is the creation or extraction of economic value by identifying and commercializing opportunities to deliver products or services, a process that typically requires consider",
    "able initiation and bears risk."
  ],
  "suggestions": [
    "\"Why do you think it's possible for robots?\"",
    "\"There's really no other way you do this.\".",
    "\"Yes.\"."
  ]
}
```

Download

Response headers

```
content-length: 1284
content-type: application/json
date: Sat, 04 Jul 2026 12:16:45 GMT
server: uvicorn
```

Personalized Networking Assistant

Generate conversation starters for networking events based on your interests.

 Enter Event Description

A tech meetup exploring how artificial intelligence is being applied to renewable energy grids and smart city infrastructure.

 Your Bio (a few sentences about yourself)

First-year Computer Science student interested in machine learning, with a growing interest in how AI can support sustainability efforts.

 Your Interests (comma-separated)

Urban planning, climate tech, machine learning

Press Enter to apply

Generate Conversation Starters

Quick Fact-Check

Enter a topic to fact-check:

Fact-Check

View Previous Conversations

Show History

View Feedback History

Show Feedback History

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Generate Conversation Starters

Extracted Themes

```
[
  0 : "artificial intelligence"
  1 : "technology"
  2 : "entrepreneurship"
]
```

Conversation Starters

- "Why don't you guys try this?"



- This person does not care what they see or hear here but has an aptitude for seeing things without ever looking away from it because that makes them happy!



- "We are all just like you" That's why it takes over an entire field that has long had very little impact on education or culture — not so much for your kids but because there wasn't



Quick Fact-Check

Conversation Starters

- "Why don't you guys try this?"



- This person does not care what they see or hear here but has an aptitude for seeing things without ever looking away from it because that makes them happy!



- "We are all just like you" That's why it takes over an entire field that has long had very little impact on education or culture — not so much for your kids but because there wasn't



Feedback noted.

Conversation Starters

- "Why don't you guys try this?"



Thanks for the feedback!

- This person does not care what they see or hear here but has an aptitude for seeing things without ever looking away from it because that makes them happy!



- "We are all just like you" That's why it takes over an entire field that has long had very little impact on education or culture — not so much for your kids but because there wasn't



 Quick Fact-Check

Enter a topic to fact-check:

Urban Planning

Fact-Check

Urban planning is the process of developing and designing plans for land use and the built environment, including air, water, and the infrastructure passing into and out of urban areas, such as transportation, communications, and distribution networks, and their accessibility. Traditionally, urban planning followed a top-down approach in master planning the physical layout of human settlements. The primary concern was the public welfare, which included considerations of efficiency, sanitation, protection and use of the environment, as well as taking account of effects of the master plans on the social and economic activities. Over time, urban planning has adopted a focus on the social and environmental "bottom lines" that focuses on using planning as a tool to improve the health and well-being of people and maintain sustainability standards. In the mid-20th century, urban planning experts such as Jane Jacobs called on urban planners to take resident experiences and needs more into consideration.

View Feedback History

Show Feedback History

 "Why don't you guys try this?"

 2026-07-04T17:51:50.536592

 "We are all just like you" That's why it takes over an entire field that has long had very little impact on education or culture — not so much for your kids but because there wasn't

 2026-07-04T17:51:34.290216

 This person does not care what they see or hear here but has an aptitude for seeing things without ever looking away from it because that makes them happy!

 2026-07-04T17:51:33.283337

 We're getting smarter through automation because we need robots around more frequently now than ever before .

 2026-07-04T16:16:46.997614

 View Previous Conversations [Show History](#) 2026-07-04T17:49:41.703789

Event: A tech meetup exploring how artificial intelligence is being applied to renewable energy grids and smart city infrastructure.

Interests: Urban planning, climate tech, machine learning

Topics: artificial intelligence, technology, entrepreneurship

Suggestions:

- "Why don't you guys try this?"
- This person does not care what they see or hear here but has an aptitude for seeing things without ever looking away from it because that makes them happy!
- "We are all just like you" That's why it takes over an entire field that has long had very little impact on education or culture — not so much for your kids but because there wasn't